

Plug and Play with Handspring™

Technology Overview

Robert Gosselin
Developer Technical Support, Handspring,Inc.

3Com
More connected.

Agenda

PalmSource 99

- Handspring's first handheld computer: the Visor™
- Developing for Handspring's handheld computer
- Springboard™ expansion slot
  - Mechanical & electrical specifications
  - Software API
  - Software considerations
- Q & A

Visor Handheld

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- Differentiation from other Palm OS® platforms
  - New applications
    - » CityTime
    - » DateBook+
    - » Advanced Calculator (MathLib is included in ROM)
  - USB connection to desktop (serial is optional)
  - Out-of-box support for PC and Mac
  - Integrated microphone
- Springboard expansion slot
  - Open mechanical architecture
  - Extension of the CPU bus

Visor Handheld Models

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- Visor Solo — *$149*
  - Case color — Graphite
  - Memory — 2 MB
  - Cradle — None
  - Software — TBD
- Visor — *$179*
  - Case color — Graphite
  - Memory — 2 MB
  - Cradle — USB
  - Software — Desktop for PC & Mac
- Visor Deluxe — *$249*
  - Case color — Graphite, ice, blue, green or orange
  - Memory — 8 MB
  - Cradle — USB
  - Software — Desktop for PC & Mac
  - Accessories — Leather case

Developing for Handspring's Handheld Computer

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- What you need to know
  - Palm OS® 3.1-compatible platform with added support for
    - » USB
    - » Springboard expansion slot
    - » Euro $ (€) character
  - True plug & play implementation
    - » Some special considerations...
- Designed for hardware expandability
- Can also be used for software distribution
  - Recommended vendor for manufacturing application-only modules

Developing for Handspring's Handheld Computer *(continued)*

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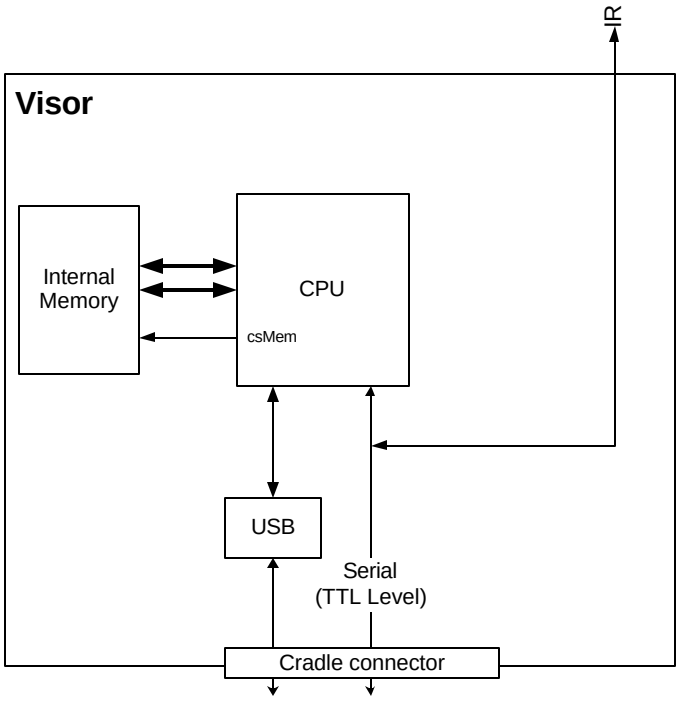
- You also need to know about Handspring's API extensions, but only if you are developing Springboard module

HsAppEventHandlerSet	Register event handler procedure
HsAppEventPost	Enable event posting by application event handler
HsCardAttrGet	Retrieve attribute
HsCardAttrSet	Set attribute
HsCardErrTry & HsCardErrCatch	Enable safe recovery for module removals
HsCardEventPost	Queue an event
HsCardPatchInstall	Enable patch installs
HsCardPatchPrevProc	Get address of previous system trap implementation
HsCardPatchRemove	Remove setup application patch
HsDatabaseCopy	Copy Palm OS® database
HsEvtResetAutoOffTimer	Reset auto-off timer

Details are available on our website in the Handspring Development Kit manual

Visor Hardware Details

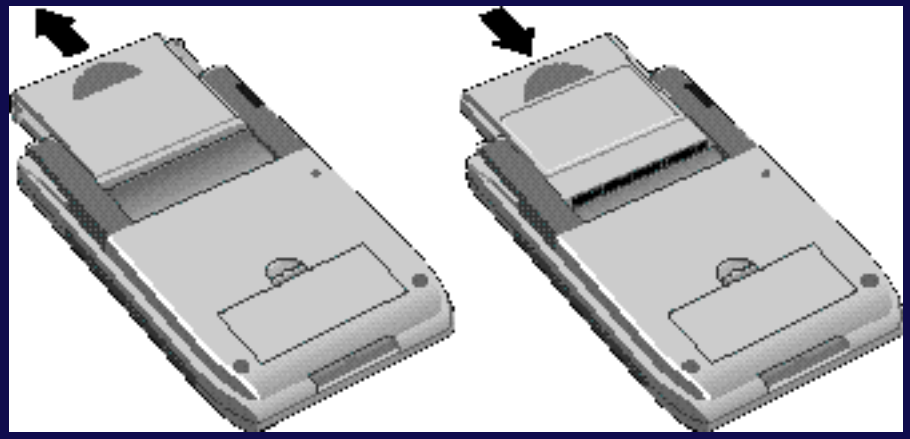
- DragonBall-EZ CPU (16.58 MHz)
- Combined USB & serial port
- Standard Palm OS® infrared
- Microphone wired directly to the Springboard expansion slot
- LCD
  - 160 x160 resolution
  - 2.2” x 2.2” (3.1” diagonal)
- Internal memory
  - 2 MB or 8 MB
- Springboard expansion slot memory
  - Maximum 32 MB



Springboard Mechanical Details

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- Slot is open on back and top to enable larger modules
- Development Kit includes all the mechanical information needed to design modules



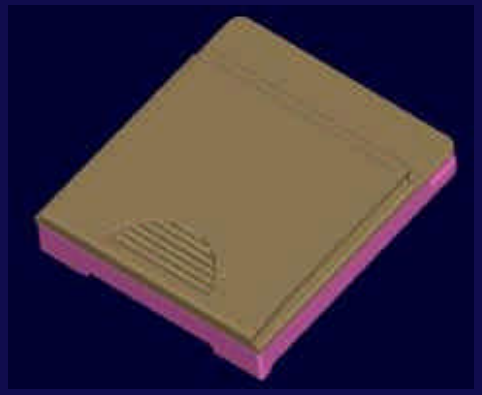
- Recommended vendor for Springboard module parts
  - ATL: Tony Sass (tonys@atllink.com), 801-434-0974, www.atllink.com
- Recommended design house for module plastics
  - IMPACC: Jay Anastas (jay@impacc.com) or Rick Mavrakis (rick@impacc.com), 650-365-7997, www.impacc.com.

Springboard Mechanical Details *(continued)*

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- Handspring offers five versions of Springboard module plastics

Springboard Standard Module

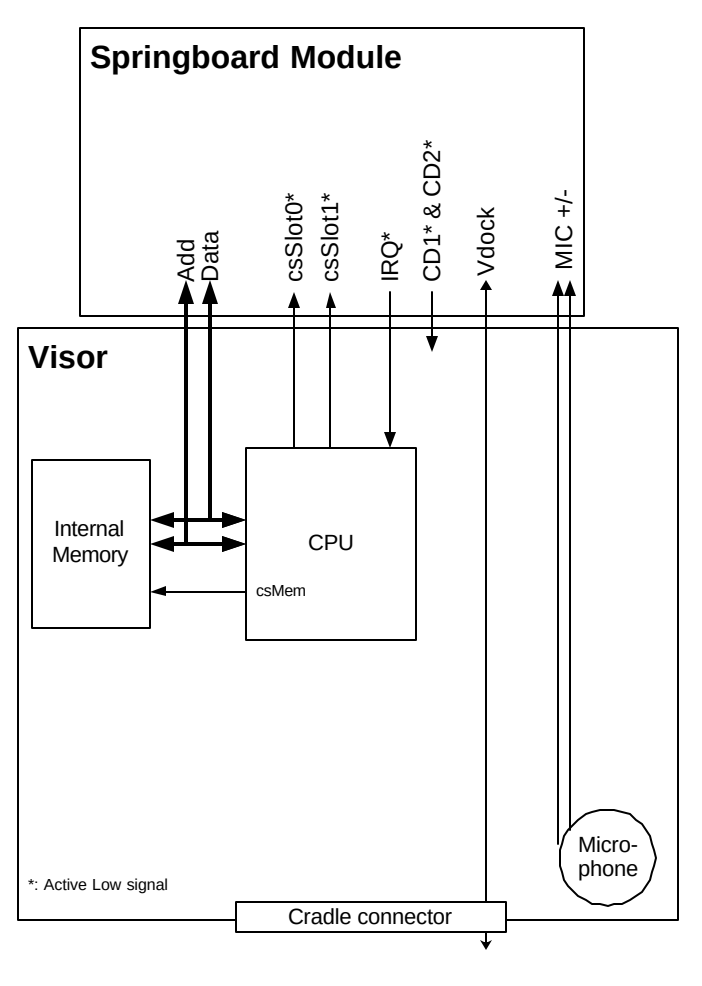


Springboard Battery Modules (00-03)



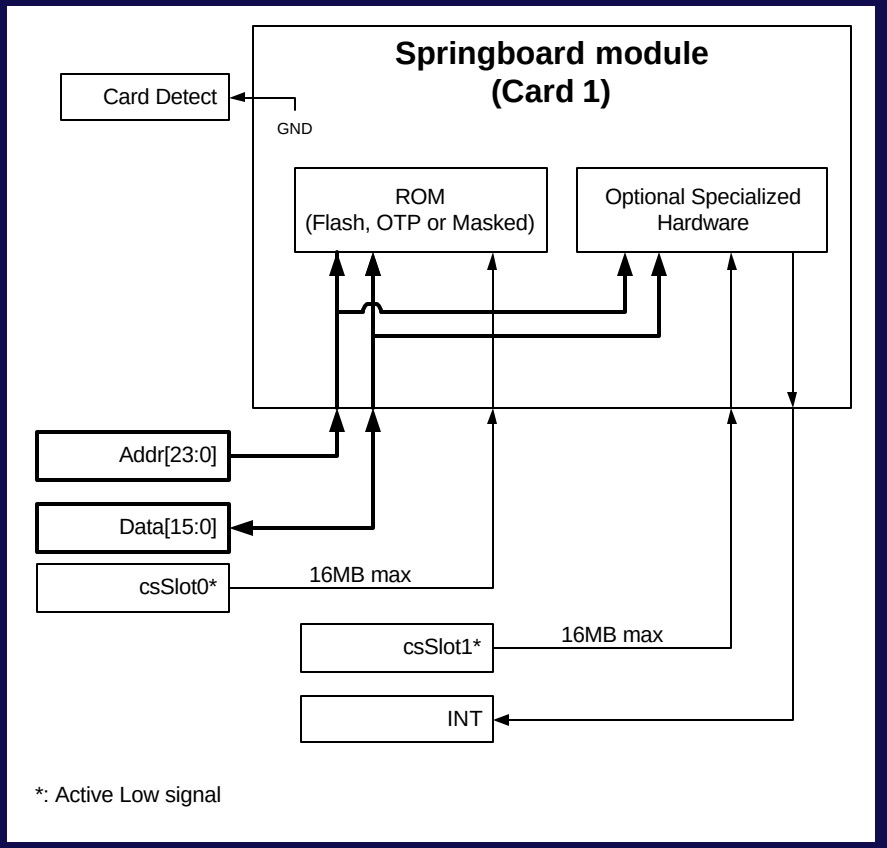
Springboard Interface Details

- Extension of the CPU bus (memory-mapped)
  - 24 address lines
  - 16-bit data bus
  - Control signals
    - » CS1*, CS2*, OE*, WE*, Int*, Reset*
  - Card detection
    - » CD1*, CD2*
  - Others
    - » VDOCK (module charging power)
    - » Mic+, Mic-
  - Power
    - » VCC: 100 mA maximum @ 3.3V
    - » GND
- ➔ *Check your power consumption!*



Springboard Module Components

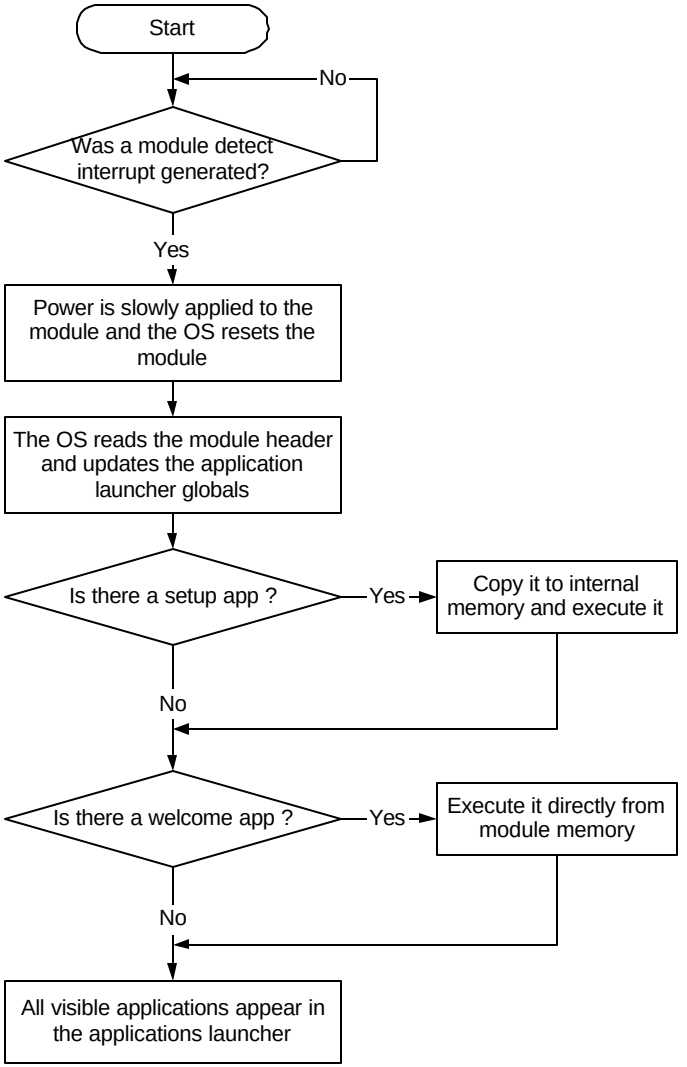
- ROM
  - Contains module header information
  - Any other applications needed for the module
  - Created with “Palm™ device makeROM” utility included in the SDK
  - Connected to csSlot0*
- Any special hardware
  - Connected to csSlot1*



Inserting a Springboard Module

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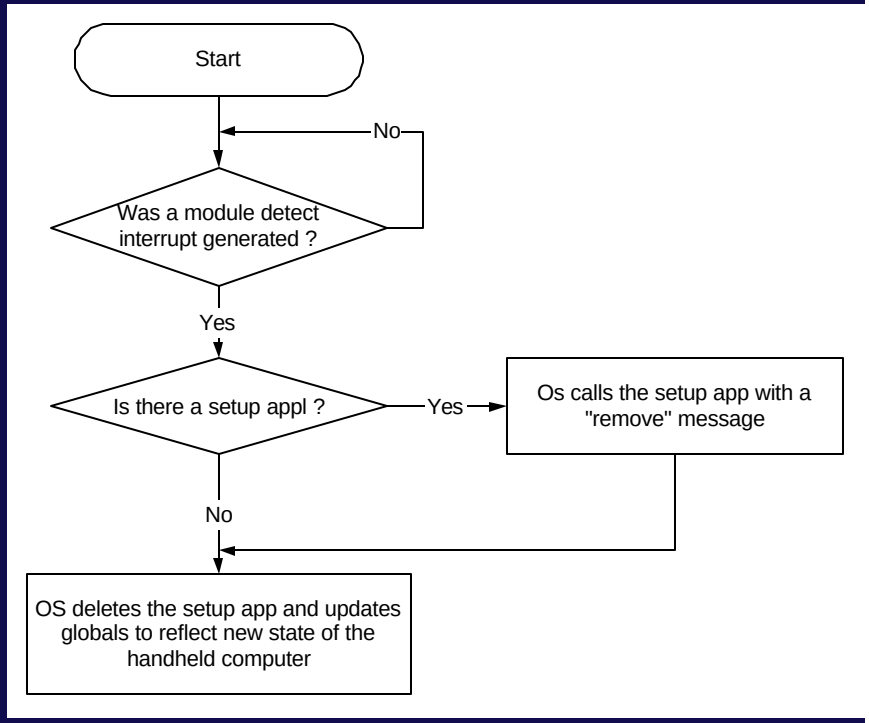
- What happens?
  - Interrupt is sent to the CPU
  - H/W powers-up the module slowly
  - OS maps the chip select signals to the default address range
  - OS resets the module
  - OS checks the card header; if valid, updates globals
  - If “setup app” is present, copy it to internal memory and execute it
  - When finished executing the “setup app”, if “welcome app” is present, execute it from module memory
  - Return to Launcher



Removing a Springboard Module

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- What happens?
  - Interrupt is sent to the CPU or a bus error occurs
  - OS “de-maps” chip select signals
  - If “setup app” is present, OS executes it with a “removal” message
  - OS deletes the “setup app” and updates globals
  - Return to Launcher (if needed)



Recommended Application Development Process

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- Design your application as before
- Test it in internal memory
- Test it in module memory using an 8MB Flash Module and its “File Mover” app to transfer your application on the module.
- If you want to sell it on a module:
  - Create a ROM image using Palm™ device MakeROM
    - » flash the memory module using Palm-Debugger
    - » send your image to recommended vendor
  - Recommended vendor for application-only module:
    - » Smart Modular, Dave Floriani, 510-624-8137, dave.floriani@smartm.com, www.smartm.com

Application Development Considerations

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- When your application is running on a Springboard-compatible module
  - The handheld's internal memory is defined as Card 0
  - The Springboard module's memory is on Card 1
    - » Opening & closing databases…
    - » or when using any calls that has the parameter "CardNo"
  - The memory on the module could be read-only
    - » Flash module
    - » OTP (One-Time Programmable) module
    - » Masked ROM module
      - – API calls like MemPtrRecoverHandle() don't work when use on ROM.

Application Development Considerations *(continued)*

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- If you are saving status information or preferences
  - The preferred way to save status information or preferences is to use the standard Palm OS® call PrefSetAppPreferences()
  - If you must use your own database, follow these steps
    - » Try writing to the card # your application is running on
    - » If it fails, try writing to Card 0

Application Development Considerations *(continued)*

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- How to handle the removal of a Springboard module while your application is running
  - Write a “valid bit” as a last step when saving a record
  - When opening a record, verify that it’s valid (if it’s a database, make sure that it works). If it isn’t valid, don’t crash, simply advise the user if necessary...
  - Try to save your “preferences” or status information as often as needed, especially after changes!

What's Available Today

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- Recommended vendors for module parts
  - Springboard module plastics: 5 versions!
  - Connectors for modules or bottom (docking) accessories
- Recommended design house for module plastics
- Recommended vendor for building application-only modules
- Complete development kit
  - Mechanical specifications
  - Electrical specifications
  - Springboard-compatible logo
  - GNU tools: compiler & linker
  - Palm debugger for source level debugging on the PC
  - Visor support in Metrowerks CodeWarrior for Palm OS® technology

What's Available Today *(continued)*

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- 8 MB Flash Module for application development
- Two subscription programs for developers
  - No-cost program
    - » E-mail support for technical questions, but no guaranteed response time
    - » E-mail support for marketing questions
    - » Access to Handspring's Knowledge base, including FAQs and technical notes
    - » Opportunities to take advantage of Handspring's marketing programs
  - Premier program (3000$/year)
    - » No-Cost program plus:
      - + Guaranteed response time (1 business day)
      - + 20% discount on the purchase of Visor and/or Visor Deluxe, with a maximum of 20 total units per year
      - + 20% discount on the purchase of select Handspring accessories

What's Coming Soon

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- Diagnostic Module (Hardware Reference Design)
  - Flash memory
  - Uses the "Battery Module" plastics
  - Test register for Read/Write
  - Sample code (already in the SDK)
  - UART (16C560)
  - LED
  - Interrupt button
- Palm OS® Emulator support for Visor
- Handspring module for the Palm OS® class
- Training module on how to use the tools
- More sample code and documentation
  - For example, an SDK for the USB port
- Sync-up forum to discuss issues & new features

Conclusions

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- If you know how to program for Palm OS® technology, you know how to program for Visor
- All the information you need is on the Web
- For technical questions
  - On the Springboard expansion slot and/or Handspring's API extensions
    - » Use the Web form ⇒ you will get your answer faster
    - » Send an e-mail to DevSupport@handspring.com
  - On Palm OS® technology
    - » Use the standard Palm Developer Technical Support Group and their website
- Subscribe to one of the developer programs – it will help you and us

All information @ www.handspring.com

Q&A


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